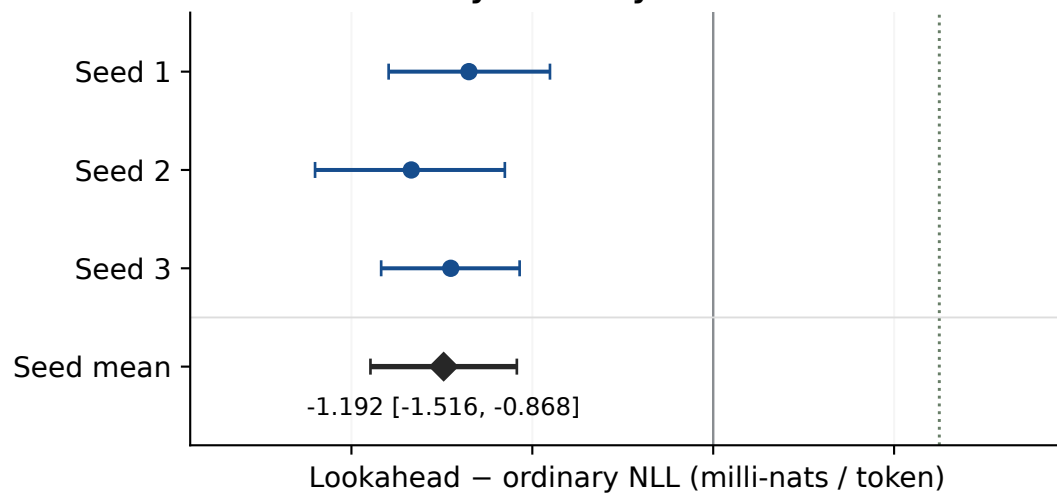


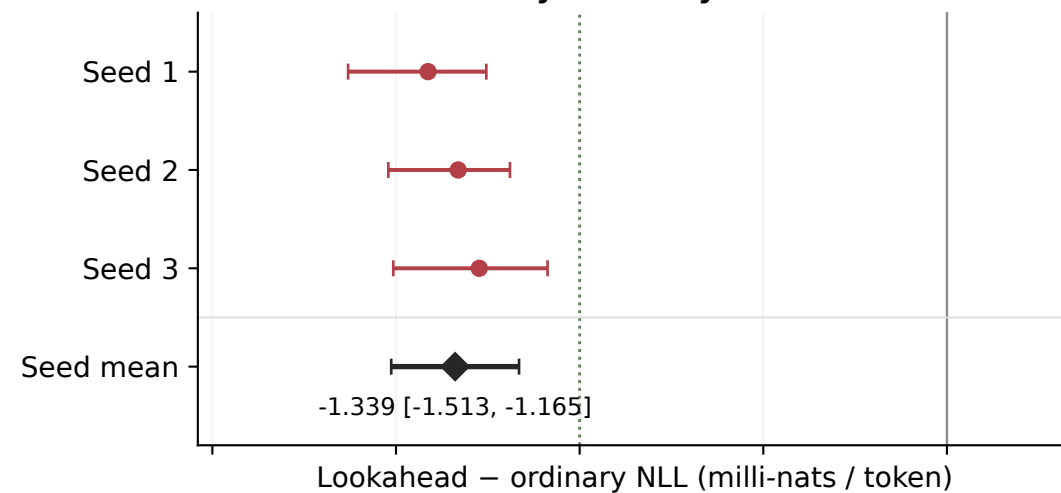
Lookahead vs refined ordinary CPT: three paired training seeds

SmolLM2-135M · four target-data passes · 20% replay · ordinary LR 8.75e-6 fixed before training

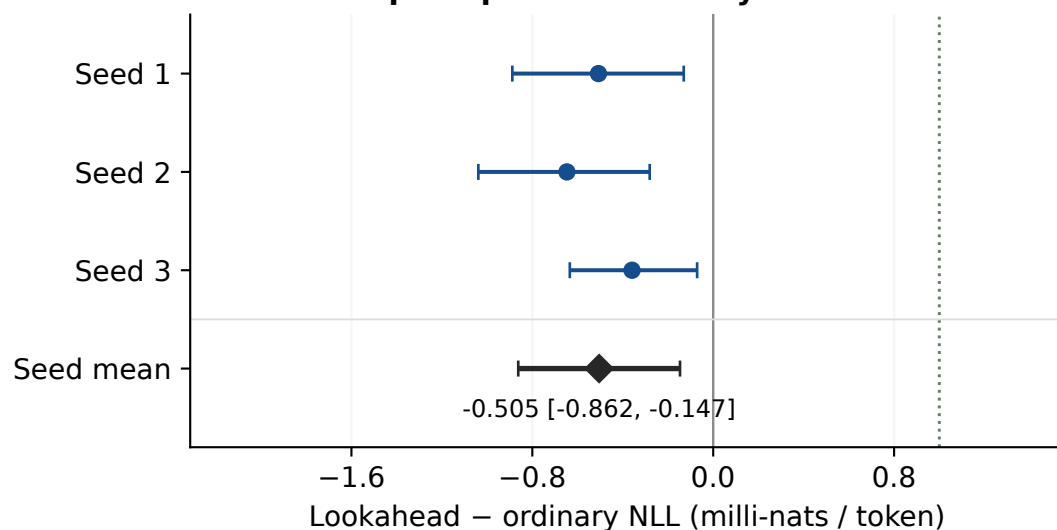
A Math · Primary: ordinary at 1664



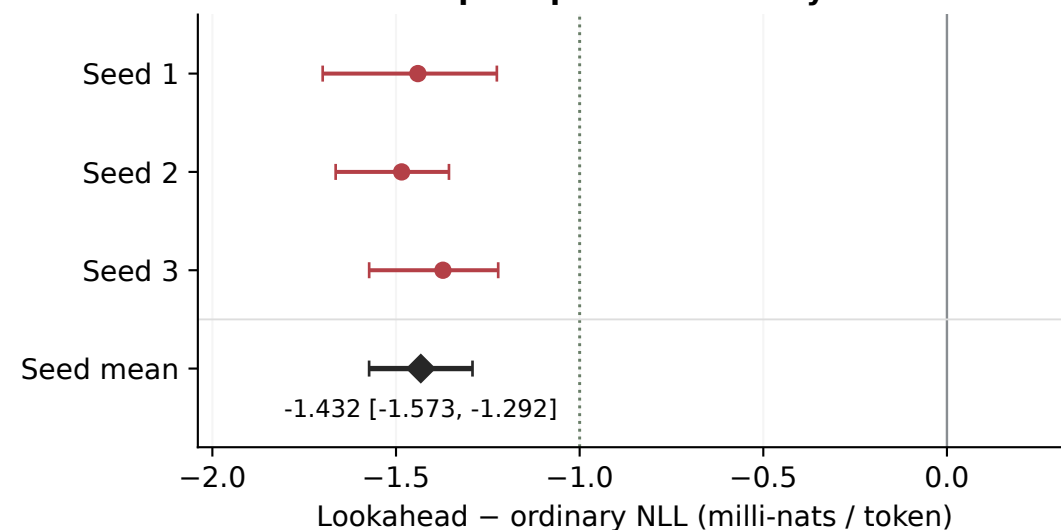
B FineWeb · Primary: ordinary at 1664



C Math · Equal updates: ordinary at 2048



D FineWeb · Equal updates: ordinary at 2048



- Individual seed: paired-document 95% interval
- ◆ Seed mean: 95% t interval (3 seeds, df = 2)

Frozen replication criteria: passed. Seed 0 excluded; previously frozen documents reused.

Candidate 2048 vs primary 1664 updates (+23.1%); equal-update results are secondary, not a wall-clock efficiency claim.

Known Lookahead-style interpolation; normalized T³ wave 2a remains null. Threshold lines: math +1 / retention –1.